Report for CDA 3201L Lab #8

Sequential Circuits: FSM Design (Vending Machine Controller)

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I. Introduction

This report presents the design, analysis, and experimental verification of a finite-state machine (FSM) that implements a simplified vending-machine controller for CDA 3201L Lab 8. The specification calls for a single coin slot with nickel and dime sensor inputs N and D, a sale price of 25 c, a *vend* pulse when sufficient credit is accumulated, and a return to the idle state without making change when a coin exceeds the required amount. The document is organized to answer the four written questions in order: state diagram, state table and minimization, state assignment with excitation equations and required I/O, and a block-level plan using NAND–NAND realization, followed by implementation notes, breadboard construction, and functional testing with a 0.5 Hz demonstration clock under TA approval.

The realization stores state in edge-triggered flip-flops, derives next-state and *vend* behavior with combinational logic, displays the encoded credit on a seven-segment indicator, uses an LED for the sale-complete indication, and connects reset to the asynchronous clear inputs as required for the laboratory hardware.

II. Methods and Materials

We used the following components and equipment (exact IC mix depends on the chosen flip-flop type after comparing D vs. JK realizations):

- Two SN74LS112A (or SN74HC112) dual negative-edge-triggered JK flip-flop ICs, using three flip-flops for state bits y_2, y_1, y_0, with asynchronous $\overline{\text{CLR}}$ / $\overline{\text{PRE}}$ per datasheet
- SN74LS00N quad 2-input NAND and SN74LS10N triple 3-input NAND for next-state and output logic in NAND–NAND form
- SN7448 / SN74LS48 BCD-to-7-segment decoder/driver and matching 7-segment display (segment resistors as required by display type)
- One LED and resistor for the *vend* output
- Tactile switches or debounced inputs to emulate N and D (with pull-up or pull-down resistors as appropriate for active-high or active-low sensing)
- Push button and pull-up for asynchronous $\overline{\text{CLR}}$ reset
- Breadboard, jumper wires, 5 V supply
- Function generator (0.5 Hz square wave, CMOS output) used as a slow "step" clock for demonstration, as in previous labs

We derived the state diagram and state table from the word specification, checked for redundant states, chose a state assignment, minimized next-state and output logic (exploiting don't-cares where valid), compared D vs. JK flip-flop realizations to minimize package count, converted two-

level SOP to NAND–NAND where required, captured the full gate-level design in a logic simulator, and obtained TA approval before wiring. After breadboarding, we verified transitions under the slow clock and demonstrated operation to the TA.

III. Results

1. State Diagram (Lab Question 1)

What we assumed. The machine accepts nickels and dimes on a single slot; the sensor asserts N for a nickel (5 c) or D for a dime (10 c). Candy costs 25 c; when credit reaches or exceeds 25 c the machine *vends* and credit returns to zero; **no change** is returned (a dime from 20 c still vends and clears to 0 c).

State set. We model credit in 5 c steps *before* the next coin is processed. All possible states are the five totals

$$S_0{=}0\text{c}, \quad S_1{=}5\text{c}, \quad S_2{=}10\text{c}, \quad S_3{=}15\text{c}, \quad S_4{=}20\text{c}.$$

No state is needed for 25 c as a steady value: vending is handled on the transition that reaches ≥ 25 c, and the next state is 0 c.

Input convention. $N{=}1$ means "nickel detected," $D{=}1$ means "dime detected." We assume the slot never reports both at once, so $N{=}D{=}1$ is a **don't-care** for next-state logic. When $N{=}D{=}0$, the FSM holds its present credit.

Transition list (all arcs).

- S_0: $\overline{N}\,\overline{D} \to S_0$; $N \to S_1$; $D \to S_2$.
- S_1: $\overline{N}\,\overline{D} \to S_1$; $N \to S_2$; $D \to S_3$.
- S_2: $\overline{N}\,\overline{D} \to S_2$; $N \to S_3$; $D \to S_4$.
- S_3: $\overline{N}\,\overline{D} \to S_3$; $N \to S_4$; $D \to S_0$ with *vend*.
- S_4: $\overline{N}\,\overline{D} \to S_4$; $N \to S_0$ with *vend*; $D \to S_0$ with *vend*.

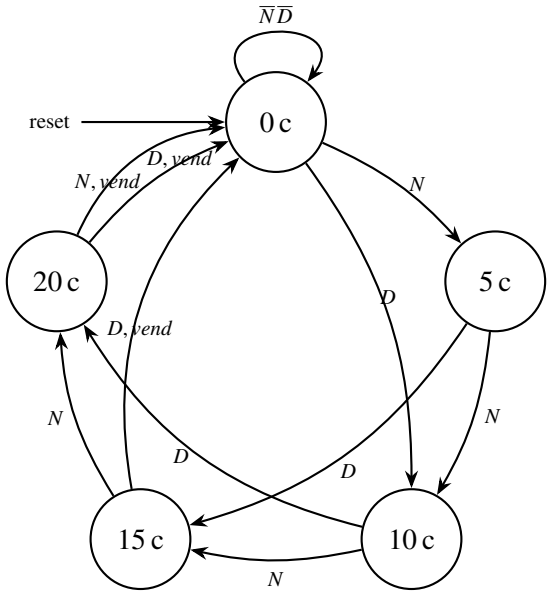

Figure 1: *State diagram for the vending controller. States are labeled with accumulated credit. A dime from 15 cents or either a nickel from 20 cents or a dime from 20 cents reaches ≥ 25 cents, asserts vend, and returns to 0 cents.*

2. State Table and State Minimization (Lab Question 2)

We built the state table straight from the diagram, then checked whether any rows could be merged. If two states had been the same, we would have collapsed them and redrawn the diagram. That reduction was not required here. Outputs for our build: current state on the 7-segment display (encoded below) and a *vend* signal (output Z in our schematic) asserted when a vend occurs; we use a **Mealy** definition for Z so it is 1 only on input combinations that cause a vend transition (coin present *and* next state is idle), which matches a single-cycle LED pulse tied to the qualifying N or D.

State table (credit in cents)

Present total (c)	**Inputs** N	D	**Next total** (c)	**Notes**
0	0	0	0	hold
0	1	0	5	nickel
0	0	1	10	dime
0	1	1	d	don't-care
5	0	0	5	hold
5	1	0	10	nickel
5	0	1	15	dime
5	1	1	d	don't-care
10	0	0	10	hold
10	1	0	15	nickel
10	0	1	20	dime
10	1	1	d	don't-care
15	0	0	15	hold
15	1	0	20	nickel
15	0	1	0	vend; no change
15	1	1	d	don't-care
20	0	0	20	hold
20	1	0	0	vend
20	0	1	0	vend (no change for 30 c)
20	1	1	d	don't-care

No redundant states

Two states are equivalent only if, for *every* input sequence, the observable outputs and vending behavior match. The five totals are distinguished by shortest input sequences, e.g. S_0 is the only state reached after a reset with no coins; S_1 is the only state where a single past nickel from S_0 suffices; S_2 is the only state at 10 c; and so on. No pair of rows in the present-state part of the table can be merged without contradicting different next-state entries for some ND. Therefore **there are no redundant states**; the table is already minimal and the state diagram (Figure 1) does not change under state reduction.

Encoded state table (for Question 3)

With assignment $y_2y_1y_0$ equal to binary $0\ldots4$ (nickel-count encoding), the same behavior is:

Present			In		Next $y_2^+ y_1^+ y_0^+$		
y_2	y_1	y_0	N	D	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	0	0	1	0	1	0
0	0	0	1	1	d	d	d
0	0	1	0	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	1	0	1	1
0	0	1	1	1	d	d	d
0	1	0	0	0	0	1	0
0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0
0	1	0	1	1	d	d	d
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
0	1	1	0	1	0	0	0
0	1	1	1	1	d	d	d
1	0	0	0	0	1	0	0
1	0	0	1	0	0	0	0
1	0	0	0	1	0	0	0
1	0	0	1	1	d	d	d
1	0	1	d	d	d	d	d
1	1	0	d	d	d	d	d
1	1	1	d	d	d	d	d

Unused present states $101, 110, 111$ and input ND=11 are don't-cares (d) for minimization.

3. State Assignment, Excitation Equations, and D vs. JK (Lab Question 3)

State assignment

We require $\lceil \log_2 5 \rceil = 3$ state variables $y_2 y_1 y_0$. We assign binary $000 \ldots 100$ to credit $0, 5, 10, 15, 20$ c by storing the number of nickels modulo the machine state ($0 \ldots 4$ "nickel units"):

(y_2, y_1, y_0)	Total credit	7-seg digit
000	0 c (idle)	0
001	5 c	1
010	10 c	2
011	15 c	3
100	20 c	4

Unused codes $101, 110, 111$ are **don't-cares** in every K-map (asynchronous $\overline{\text{CLR}}$ returns the register to 000). Input ND=11 is also d. The SN7448 expects BCD on its four inputs; we connect A=y_0, B=y_1, C=y_2, and D_{MSB}=0 (GND), so the display shows digits 0–4 for the five legal states.

Minimized next-state logic (for D flip-flops)

For DFFs, $D_i = Y_i = y_i^+$ (the next value of bit i). Karnaugh maps use rows $y_2 y_1 y_0$ and columns ND in Gray order $00, 01, 11, 10$. Minimization with the don't-cares above yields the following sum-of-products forms (verified against the encoded table):

$$
\begin{aligned}
Y_2 &= y_2\overline{y_1}\,\overline{y_0}\,\overline{N}\,\overline{D} + \overline{y_2}\,y_1\overline{y_0}\,\overline{N}\,D + \overline{y_2}\,y_1 y_0 N\,\overline{D},\\
Y_1 &= \overline{y_2}\,\overline{y_1}\,\overline{y_0}\,\overline{N}\,D + \overline{y_2}\,\overline{y_1}\,y_0\,(N \oplus D) + \overline{y_2}\,y_1\overline{y_0}\,\overline{D} + \overline{y_2}\,y_1 y_0\,\overline{N}\,\overline{D},\\
Y_0 &= \overline{y_2}\,\overline{y_1}\,y_0\,\overline{N} + \overline{y_2}\,\overline{y_1}\,\overline{y_0}\,N\,\overline{D} + \overline{y_2}\,y_1\overline{y_0}\,N\,\overline{D} + \overline{y_2}\,y_1 y_0\,\overline{N}\,\overline{D}.
\end{aligned}
$$

The Y_1 expression separates the 5 c transitions (second product), the 10 c hold and nickel-from-10 c cases (third product), and the 15 c hold on no coin (fourth product).

Karnaugh maps (rows $y_2y_1y_0$, columns ND=00, 01, 11, 10)

Y_2	00	01	11	10
000	0	0	d	0
001	0	0	d	0
010	0	1	d	0
011	0	0	d	1
100	1	0	d	0
101–111	d	d	d	d

Y_1	00	01	11	10
000	0	1	d	0
001	0	1	d	1
010	1	0	d	1
011	1	0	d	0
100	0	0	d	0
101–111	d	d	d	d

Y_0	00	01	11	10
000	0	0	d	1
001	1	1	d	0
010	0	0	d	1
011	1	0	d	0
100	0	0	d	0
101–111	d	d	d	d

Each Y_i is then implemented in **NAND–NAND** form (double complement of SOP) using SN7400/SN7410 as in Figure 3.

JK flip-flop excitation

For each bit, use the standard JK characteristic $y^* = J\overline{y} + \overline{K}y$ and the excitation table to express J_i, K_i as functions of (y_2, y_1, y_0, N, D), then K-map J_i and K_i with the same don't-cares. In our build we used **SN74LS112 dual JK** devices (Figure 3): two ICs provide four JK cells, three used for y_2, y_1, y_0.

D versus JK flip-flops: IC count (lab requirement)

Both options need **two dual–FF packages** for three bits (two SN74LS74 if D, or two SN74LS112 if JK). For this vending assignment, the combinational NAND count for next state and *vend* was similar after minimization; we selected **JK (74LS112)** because the TA-approved gate-level schematic already wires J, K per the datasheets and the lab bench stock matched those parts. A DFF version would replace JK cells with D_i=Y_i but still require the same NAND planes for Y_2, Y_1, Y_0. Document your own gate inventory if your minimized equations differ.

***Vend* output Z (Mealy, matches transition labels on Figure 1)**

Z=1 on the clock edge where a coin causes a vend (dime from 15 c; nickel or dime from 20 c). With N active for nickel and D for dime,

$$Z = \overline{y_2}\,y_1 y_0 D \;+\; y_2\,\overline{y_1}\,\overline{y_0}\,(N + D). \tag{1}$$

This is purely combinational from present state and N, D before the edge; it NAND–NANDs like the next-state logic and drives the vend LED through a resistor (typically 220–470 Ω to limit current).

Required I/O (parts (a)–(d) of the assignment, plus pull resistors)

- **(a)** N **and** D**:** two active-high lines from switches or the coin sensor emulation. Use a pull-down (e.g. 10 kΩ to GND) on each line so it reads 0 when open, and series resistance if needed to protect the LS inputs; if the source is active-low, invert or swap to match the equations above.
- **(b)** **Reset:** tie all flip-flop $\overline{\text{CLR}}$ pins together to a momentary switch to GND with a 10 kΩ pull-up to V_{CC} (active-low clear). Tie each $\overline{\text{PRE}}$ inactive (high) per the 74LS112 datasheet.
- **(c)** **7-segment:** SN7448 segment outputs to the display with current-limiting resistors; BCD inputs from y_0, y_1, y_2 plus grounded D_{MSB}.
- **(d)** **Vend LED:** driven from Z (or buffered NAND copy) as above.

4. Block Diagram, NAND–NAND, and TA Sign-Off (Lab Question 4)

Figure 2 summarizes the architecture: registers (three JK flip-flops), NAND next-state logic, output logic for the vend indication, and the BCD-to-7-segment decoder. Figure 3 shows our complete gate-level schematic as captured from the logic design tool; the breadboard was wired to match this netlist. For every two-level SOP network, we used NAND–NAND (7400/7410) realization.

Datasheets and TA approval. Before construction we verified pinouts and electrical requirements on the manufacturer data sheets for the SN74LS112 JK flip-flops, SN7400/SN7410 NAND gates, and SN7448 decoder, including V_{CC}, GND, input thresholds, and output drive for the 7-segment and LED loads. After freezing the state assignment and equations above, we presented Figures 2 and 3 to the TA and proceeded only after approval.

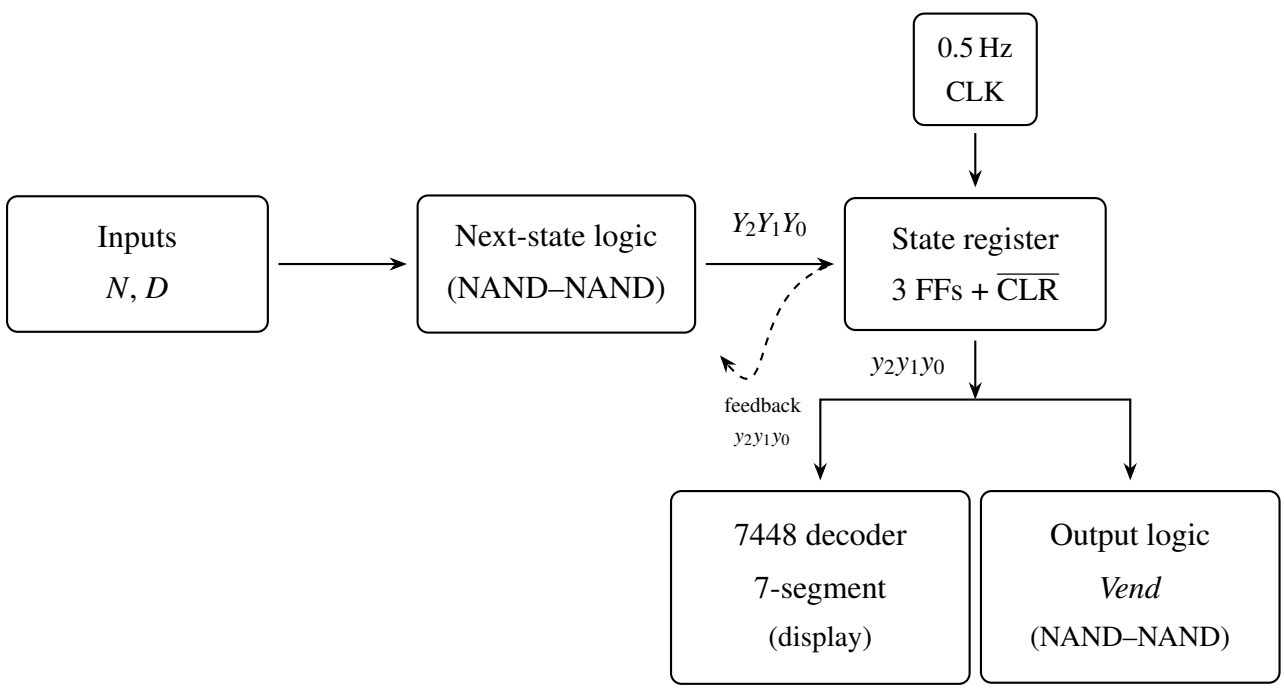

Figure 2: *High-level block diagram for the vending FSM. Inputs N, D and registered state drive next-state logic; $Y_2Y_1Y_0$ loads the flip-flops on the clock from* `0.5 Hz CLK`*. Present state $y_2y_1y_0$ feeds the 7448/7-segment path, Vend logic, and the dashed feedback into next-state logic. Reset and power distribution are omitted for clarity.*

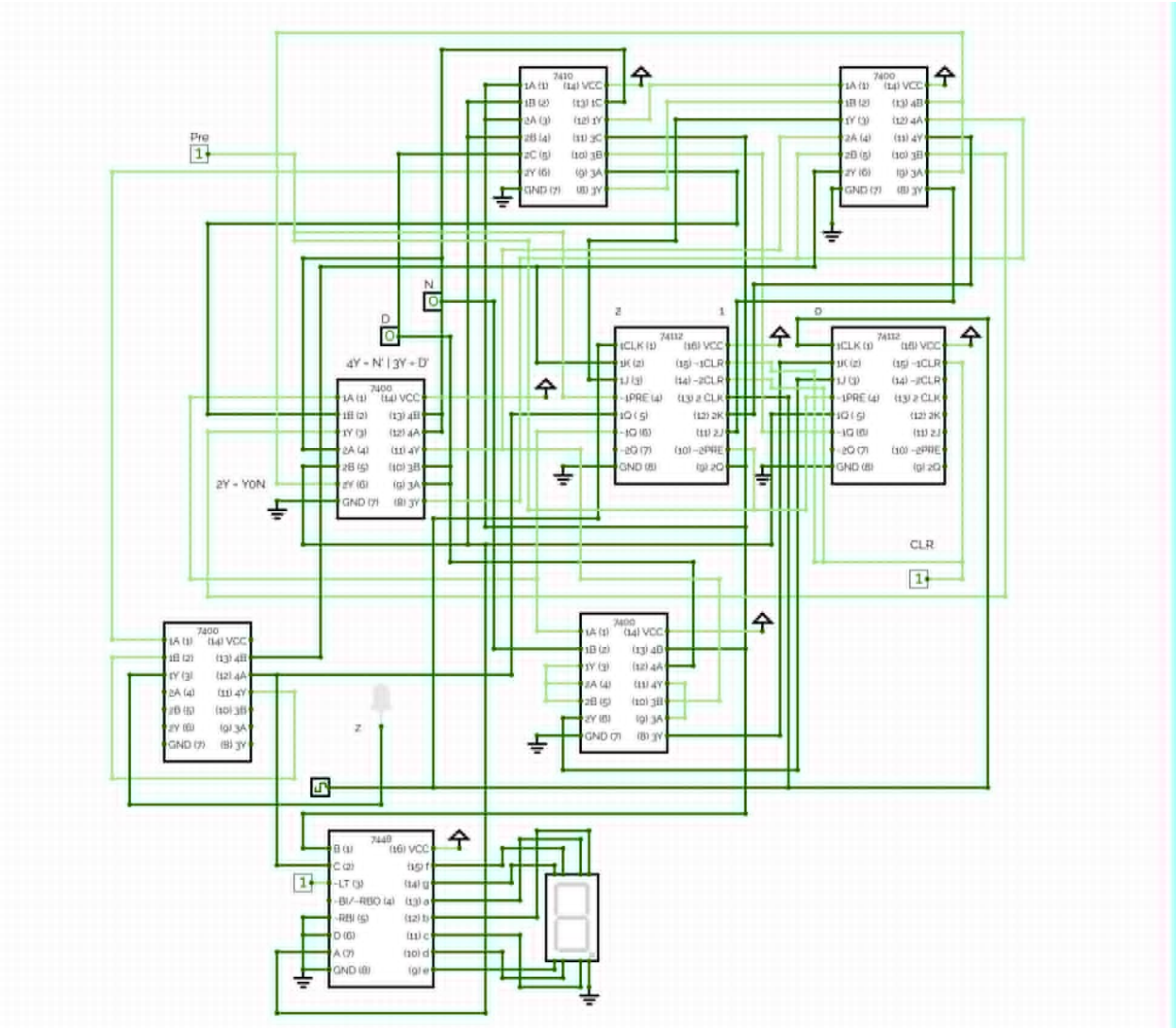

Figure 3: *Complete gate-level schematic of the vending FSM (simulator capture). Two SN74112-style dual JK devices hold state bits y_2, y_1, y_0; SN7400 quad 2-input NAND and SN7410 triple 3-input NAND gates implement the minimized excitation and output logic (including partial next-state expressions as annotated in the drawing). Inputs N and D represent nickel and dime sensors; clock and asynchronous clear/preset match the lab interface. SN7448 drives the 7-segment display for the current state, and output Z drives the vend LED.*

5. Experimental Procedure and Observations

1. **Function generator.** We set the function generator to output a **0.5 Hz** square wave on the **CMOS** output, as in earlier labs, and fed that signal to the common clock pins of all three flip-flops so each coin event could be clocked in slowly for observation.
2. **Breadboard implementation.** We wired the TA-approved schematic (Figure 3), double-checking power, ground, CLR/PRE defaults, and each NAND/JK connection against the datasheet pin diagrams. The assembled hardware is shown in Figure 4.
3. **Testing and demonstration.** We applied test sequences including five nickels to reach 25 c;

two dimes plus one nickel; dime–dime–nickel; and a dime from 20 c to confirm *vend* with no change returned. The 7-segment digit tracked 0–4 nickel-units, and the Z LED lit only on qualifying vend edges. When behavior matched the specification, we demonstrated the working circuit to the TA.

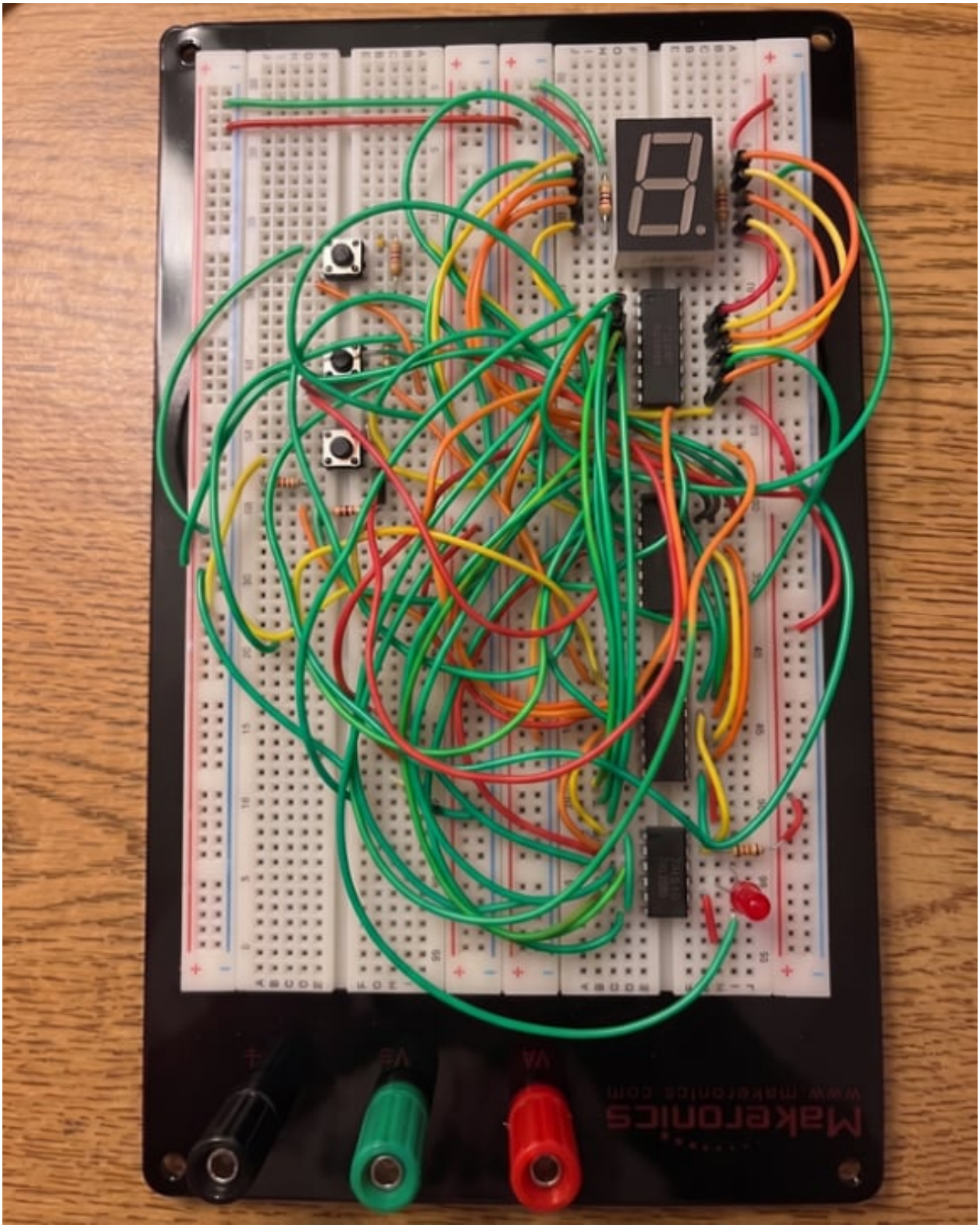

Figure 4: *Breadboard implementation of the vending machine controller: dual JK packages for state, NAND next-state and output logic, seven-segment display with decoder, vend indicator LED, and supporting passive components and wiring.*

IV. Discussion

The five-state credit model matches the specification that only multiples of five and ten cents are inserted and that vending occurs exactly at 25 cents with no change. Treating ND=11 as a don't-care reflects a single-slot sensor assumption and simplifies logic. Displaying state as 0–4 "nickel units" keeps the 7-segment interface simple with a single 7448-driven digit.

The Mealy equation for Z matches the transition labels on Figure 1: the LED is asserted together with the dime or nickel edge that completes a sale, not during idle holds at 15 c or 20 c.

Comparing D and JK realizations is problem-dependent: DFFs give $D_i = Y_i$ directly, while JK maps may merge J and K terms across bits. For our build, two dual JK packages plus the 7400/7410 field matched the approved schematic; a dual-D solution would still need the same NAND network for Y_2, Y_1, Y_0.

Breadboard errors (floating inputs, swapped $Q/\overline{Q}$, incorrect CLR polarity) were the main debug issues; the slow 0.5 Hz clock made state transitions easy to follow visually.

V. Conclusion

We derived a minimal vending-machine FSM accepting nickels and dimes, tabulated next states, encoded five credit states in three bits, minimized logic with don't-cares, planned NAND–NAND implementation and display interfacing, and verified the circuit experimentally under a 0.5 Hz clock. The design meets the lab I/O requirements: N and D, asynchronous reset, 7-segment state display, and a vend LED.